

Predictive Insight Engine

Part A: Theory

1. What are Supervised Learning Algorithms?

- Supervised Learning algorithms learn from labeled data (input and correct output are already known).
- They identify patterns and make predictions on new data.
- Used for both regression and classification problems.
- Example: Predicting house prices using previous house data.

2. Difference Between Regression and Classification Algorithms

Regression

- Predicts numerical values.
- Output is a number.
- Example: Predicting a house price.

Classification

- Predicts categories or classes.
- Output is a label.
- Example: Predicting whether an email is Spam or Not Spam.

3. Explain Simple Linear Regression

- Simple Linear Regression predicts a value using one input variable.
- It shows the relationship between input and output using a straight line.
- Used when one factor affects another.
- Example: Predicting house price based on house size.

Equation:

$$y = mx + c$$

4. Assumptions of Linear Regression

1. Linearity – Input and output should have a straight-line relationship.
2. Independence – Data points should not depend on each other.
3. Homoscedasticity – Errors should be evenly distributed.
4. Normality – Errors should follow a normal distribution.
5. No Multicollinearity – Input variables should not be highly correlated.

5. What is Bias-Variance Trade-Off?

- Bias means the model is too simple and misses important patterns.
- Variance means the model is too complex and learns noise from the data.
- High bias causes underfitting, while high variance causes overfitting.
- A good model balances both for better predictions.

6. Explain Overfitting and Underfitting with Examples

Overfitting

- The model learns training data too closely, including noise.
- Performs well on training data but poorly on new data.
- Example: A very complex model that perfectly fits training house prices but fails on unseen houses.

Underfitting

- The model is too simple to learn patterns in the data.
- Performs poorly on both training and testing data.
- Example: Using a simple straight line to predict house prices when the relationship is complex.